

proved optical output of the infrared LEDs and optimised photodiode arrays in the photodetector devices, resulting in a maximum turn-on time of just 0.25ms, a 50% reduction compared to the previous model.

Toshiba Electronic Devices & Storage Corp.
<https://toshiba.semicon-storage.com>

GaN-on-Sapphire MOSFET

Transphorm has introduced the industry's first 1200 V GaN-on-Sapphire MOSFET. The GaN-on-Sapphire construction leverages the excellent insulating properties of sapphire, enabling much higher operating voltages. These GaN switches, based on high electron



mobility transistors (HEMTs) on sapphire substrates, offer fast-switching and low-loss capabilities at voltages beyond traditional GaN-on-Silicon

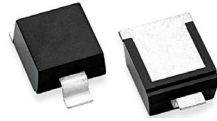
devices, which typically support up to 650V. This breakthrough MOSFET will enhance performance and safety in high-powered applications like electric vehicles (EVs) and three-phase power systems. With the growing demand for higher voltage ratings in power electronics, this 1,200V GaN-on-Sapphire device is a game-changer for EVs and high-voltage industrial applications.

Transphorm
<https://www.transphormusa.com>

High-power TVS diode

Littelfuse has introduced the LTKAK2 series of high-power TVS diodes, offering robust circuit protection for high-power applications while requiring 60-80% less PCB space. These surface mount solutions address the need for

TVS diodes compatible with automated PCB assembly processes and featuring a high surge rating. With a compact design and a modified SMT0-218 package, the LTKAK2-L series



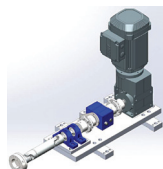
enables designers to achieve circuit protection in a smaller footprint compared

to axial leaded solutions. The diodes provide a surge rating of 2kA, meeting the requirements of IEC-61000-4-5 Level 4, making them suitable for high-power AC and DC applications across various industries including avionics, power supplies, and converters. This innovative product addresses the challenge of finding high surge rating TVS diodes in smaller-sized, automated assembly-compatible packages, allowing for more accessible and reliable circuit protection in high-power designs.

Littelfuse
<https://www.littelfuse.com>

Wireless torque sensor rigs

SGN Electrical has developed two test rigs equipped with wireless torque sensors to effectively assess the landing gear and examine the



flaps of an aircraft's tailplane. These test rigs play a crucial role in ensuring the safe and reliable landing of transoceanic-scale

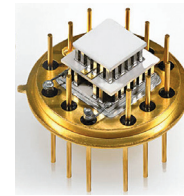
airliners responsible for transporting hundreds of passengers. The first test rig focuses on evaluating the ball screws responsible for controlling the movement of the tailplane flaps. It goes beyond functional testing by sim-

ulating the airflow experienced during high-speed flights. The second test rig is dedicated to assessing the landing gear, replicating the forces exerted on the main lead screw during landings. Both test rigs employ the TorqSense transducer, eliminating the need for slip rings and simplifying the setup process. By utilising wireless torque sensors, the testing procedure can be streamlined, allowing for more tests to be conducted within a given time frame and ensuring thorough verification for enhanced aircraft safety.

SGN Electrical
www.sgnelectrical.com

Micro-multistage thermoelectric cooler

Laird Thermal Systems has introduced the OptoTEC MSX series of micro multistage thermoelectric coolers, specifically designed for miniature



optical packages in high performance image sensing applications. These coolers, integrated into optical packages like TO-39,

TO-46, or TO-8, create advanced optical thermoelectric assemblies (TEAs). Utilising advanced ceramic materials, proprietary solder bonding techniques, and next-generation thermoelectric materials, the MSX series delivers up to 10% higher cooling capacity in a micro footprint. Deep cooling below ambient temperature is crucial for high-end image sensing devices to maximize image resolution, and the MSX series meets this requirement. The micro multistage coolers offer small footprints and high heat pump-



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